THE RELAY / Vaani ML Engineering Intern

Evidence-first pipeline • measured LOCAL stand-in • real Groq measurements unavailable

1 What I built and why

Five steps: classify, retrieve page-cited passages, scope/plan, stream a draft, and validate citations. Branch to abstention if evidence is weak. Key decision: fail closed rather than fabricate a quick answer.

2 The real service and my stand-in

Groq adapter: streaming chat completions. No Groq key was available, so real service limits, headers, and calibration are NOT MEASURED. Simulator: chosen lognormal jitter, 503 errors, minute/day admission limits, queue timeouts, load-sensitive delay and word-by-word streaming. Token figures are character-count proxies.

3 Results — every table recomputed from results.csv

TABLE 1 • Hosted Groq limits (provider documents describe 429/retry-after; account-specific values unobserved)

Limit	Value	Method	Confidence
Requests/minute	not measured	no account key	low
Tokens/minute	not measured	no account key	low
Requests/day	not measured	no account key	low
Tokens/day	not measured	no account key	low
Concurrent inflight	not measured	no account key	low
Actual 429 payload	not measured	no account key	low
Reset time	not measured	no account key	low

TABLE 2a • Internal model TTFT only (not user-visible); conditional on token emission; NM = no token

Scenario	Source	N	Duration s	Offered/min	TTFT p50 / p95 / p99 ms	Total p50 / p95 / p99 ms
steady	stand-in	12	3.38	212.8	189.9/825.7/826.2	305.9/961.9/962.1
burst	stand-in	22	0.42	3150.4	66.6/289.2/329.6	234.5/247.6/382.5
sustained_overload	stand-in	28	0.95	1776.2	137.0/400.0/440.6	0.5/420.2/500.9
rpm_halved	stand-in	16	1.05	912.9	202.8/365.5/404.2	0.6/424.5/525.4
daily_exhausted	stand-in	10	0.54	1107.8	NM/NM/NM	0.5/0.8/0.9

TABLE 2b • Outcomes (API charge is zero for local stand-in; CPU and electricity excluded)

Scenario	Succeeded	Rate limited	Timeout/error	Retries/req	Peak queue	Tokens/ans	API $/1000
steady	100.0%	0.0%	0.0%	0.00	0	119.6	$0.00
burst	27.3%	31.8%	40.9%	0.00	8	19.5	$0.00
sustained_overload	32.1%	53.6%	14.3%	0.00	4	27.4	$0.00
rpm_halved	43.8%	56.2%	0.0%	0.00	0	48.6	$0.00
daily_exhausted	0.0%	100.0%	0.0%	0.00	0	0.0	$0.00

TABLE 3 • Retrieval evaluation, 18 authored questions per setting (15 in-scope and 3 outside)

Piece/top-k/rerank	N	Correct+cited	Wrong	Correct refusal	Tokens/ans	TTFT p50 / p95	Total p50 / p95
220/1/no	18	83.3%	0.0%	16.7%	64.1	69.8/133.4	164.5/231.6
500/3/yes	18	83.3%	0.0%	16.7%	103.1	69.9/133.5	165.1/231.7
1000/6/yes	18	83.3%	0.0%	16.7%	142.1	70.0/133.7	165.3/232.0

All simulations ran against 19 paraphrased, page-verified notes; full-paper PDFs could not be downloaded in this environment. 88 load calls + 54 retrieval calls + 2 retry trials = 144 measured stand-in rows. These are NOT results for a calibrated real provider.

ENGINEERING JUDGMENT / FAILURE MODES

4 Where the time goes

Slowest successful request steady-011: total 962.16 ms; first token 826.36 ms. Direct queue wait 0.0 ms. Work after queue admission through first token 825.9 ms; work after first token through verification 135.8 ms; no retries. Evidence: monotonic timestamps in artifacts/traces.json.

5 When a limit is hit

Minute cap: reject calls with long retry-after; do not occupy queue for a minute. Daily cap: stop issuing hosted calls and return a clear, safe service-unavailable message. Prioritize requests already accepted; allow cheap evidence-based local refusals. Neither case silently swaps to speculative answers.

Retrieval: quality versus cost

All three note-based settings scored 15/18 correct+cited and 3/18 correct refusal. Small 220-character pieces / top-1 minimizes proxy tokens here. For a full 120-page source library I would retest because wider passages may be necessary. Near limits: lower top-k, preserve citations and abstention. First-token time stays similar in this simple simulator because it has no true token-dependent prefill.

How this library will mislead a naive retriever

1. "Attention" appears across the corpus: a spoken dialogue query can retrieve Vaswani rather than Moshi. 2. Moshi gives both 160 ms theoretical and 200 ms practical latency: an ambiguous latency query can cite the wrong number. 3. "Chunking" means learned segmentation in H-Net but fixed text windows in this retrieval system; a vague query conflates them.

6 Retries

Only once, on a pre-token transient 503, with a short exponential backoff; never replay partial output. Controlled failure injection: fail-fast error took 110.85 ms; retry succeeded in 595.65 ms. The retry purchased completeness but made the caller wait longer and consumed another admission. Never retry a 24-hour limit.

7 What the person sees

"Today's model-call budget is exhausted. I cannot verify an answer from these papers right now, and I will not guess. Please try again after the quota resets."

8 What I am unsure about, left out, learned

Least certain assumption: performance on curated notes predicts performance on the full papers; it does not. Left out: dense reranker and distributed queue. New lesson: TTFT on successful responses can look excellent even while most offered requests are refused; show coverage and end-to-end completion together.

Your feedback on Vaani

The Teacher AI prompt-diff review and Memories product stood out. The website highlights P90 latency around 450 ms, but the timing boundary is unclear to a first-time reader. I would publish stage-specific time definitions and a language-filterable sample of real calls so developers can judge quality versus responsiveness.

How I worked with the assistant

Accessible raw user prompt and work log: conversation_history.md. No second or third direct user prompts were made in this thread; instead these were the three governing instructions:

1. User: complete the PDF assignment and put it in the named GitHub repository. This set the deliverable scope.

2. Assignment: pin the three arXiv PDF versions and refuse unsupported factual claims. This shaped citation guards.

3. Assignment: submit raw per-request CSV and recomputable percentile tables. This forced explicit instrumentation.

The assistant initially kept status=error even after a successful retry. The forced 503 retry trial exposed this; the implementation now resets status=ok on success. One deliberate override: do not claim fabricated real Groq measurements or simulator calibration. Applicant-specific choices are NOT fabricated; the applicant must personally run Groq and describe their own genuine review/overrides before applying.

Boundary of these results

Recorded: 6 passing tests; local simulator latency distributions; 18-question × 3 retrieval evaluations; 5 load shapes; failure-injection trace. Missing: real Groq key, measurements, fitted stand-in, full-PDF benchmark. This is a substantive reproducible prototype, not yet a fully compliant final submission.